

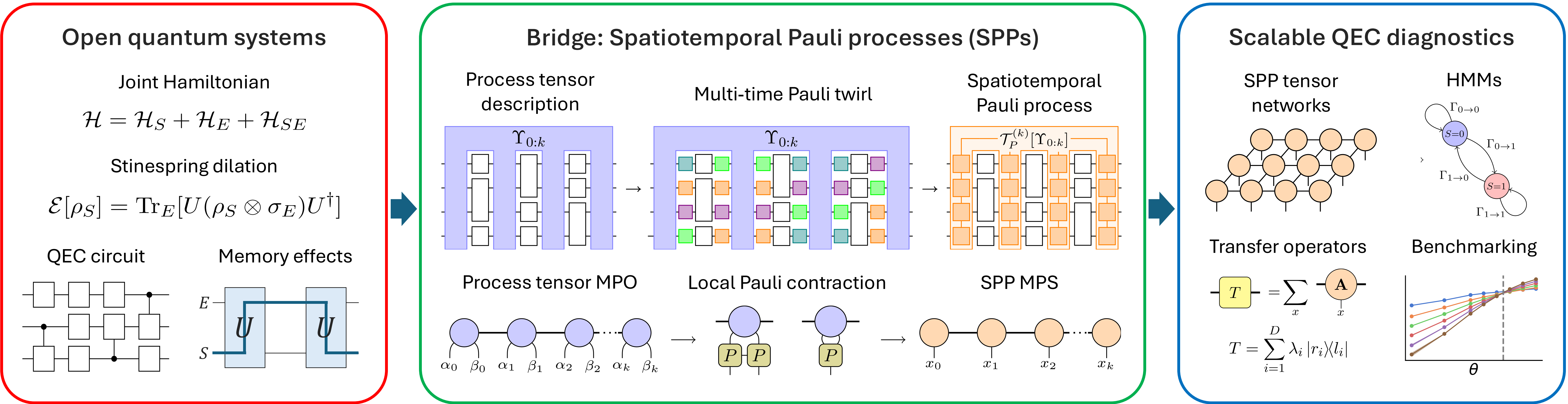
Spatiotemporal Pauli Processes: Quantum combs for modeling correlated noise in quantum error correction

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Main idea

Correlated noise is a critical QEC failure mode: temporal memory and spatial structure can concentrate faults into damaging error bursts. **Spatiotemporal Pauli Processes (SPPs)** model these effects as stochastic processes over Pauli faults across space and time. Crucially, they retain an operational bridge to microscopic non-Markovian dynamics via the multi-time Pauli twirl of a general process tensor. This framework brings tensor network structure, transfer operator diagnostics, and hidden Markov sampling into QEC-compatible noise models. We use it to show how temporal memory degrades surface code performance, and how quantum cellular automaton (QCA)-derived bath dynamics can generate pseudo-critical error avalanches that break distance scaling.

Framework overview



Resources

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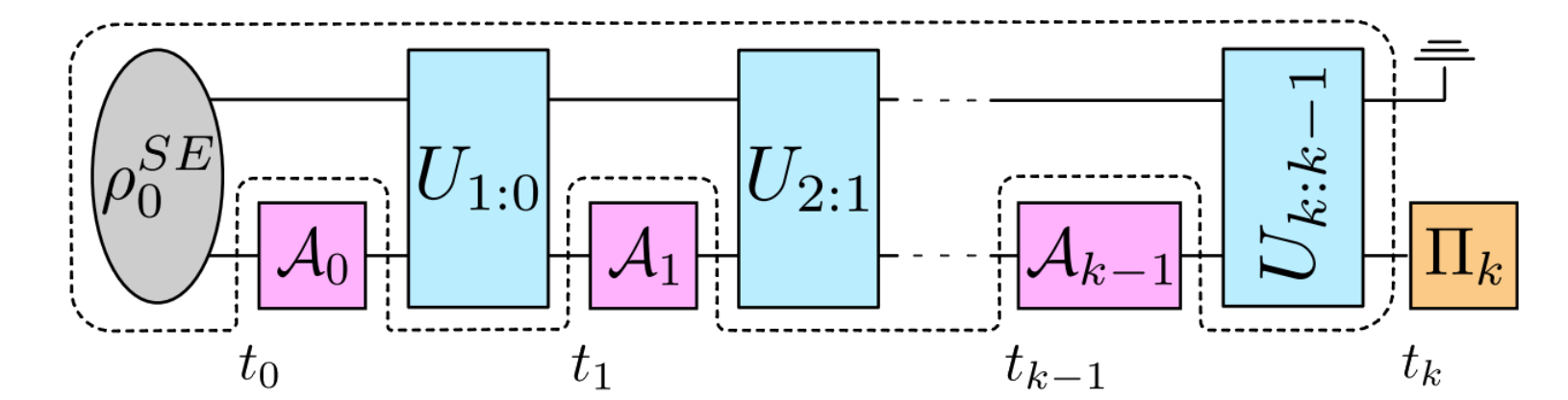


Paper



Code + Poster

Process tensors and Pauli twirling



A **process tensor** generalizes a quantum channel to many times. It encodes how an environment with memory responds to interventions,

$$\mathcal{T}_{0:k}[\rho_{\text{in}}, \mathbf{A}_{0:k-1}] = \rho_{\text{out}}.$$

We can represent the process tensor as a Choi operator (many-body quantum state),

$$\Upsilon_{0:k} \in \mathcal{B}\left(\bigotimes_{j=0}^k \mathcal{H}_j^{\text{in}} \otimes \mathcal{H}_j^{\text{out}}\right).$$

Pauli twirling projects a general quantum channel onto a Pauli channel,

$$\mathcal{T}_P(\mathcal{E})[\rho] = \frac{1}{|\mathbb{P}(n)|} \sum_{P \in \mathbb{P}(n)} P\mathcal{E}(P\rho P)P.$$

Spatiotemporal Pauli processes

Pauli twirling each input-output pair independently yields the **multi-time Pauli twirl**,

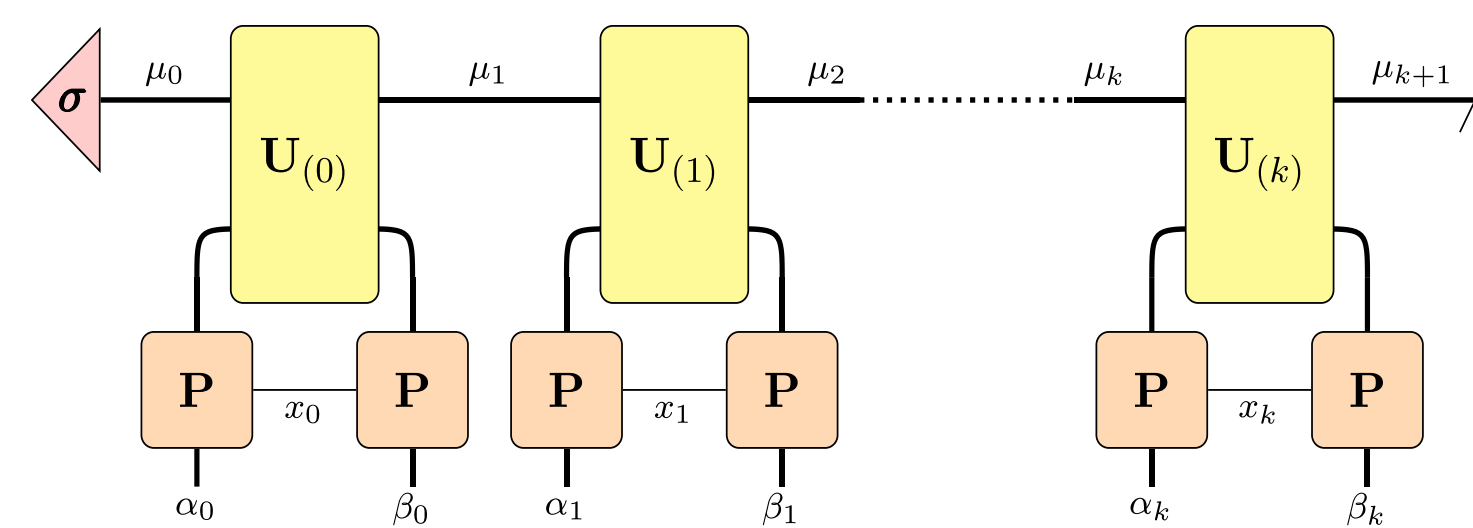
$$\mathcal{T}_P^{(k)}(\Upsilon_{0:k}) = \frac{1}{|\mathbb{P}(n,k)|} \sum_{P_0, \dots, P_k} \left(\bigotimes_{j=0}^k P_j \otimes P_j \right) \Upsilon_{0:k} \left(\bigotimes_{j=0}^k P_j \otimes P_j \right).$$

The multi-time Pauli twirl maps a general process tensor to a **spatiotemporal Pauli process** (SPP): a stochastic process over Pauli faults across space and time

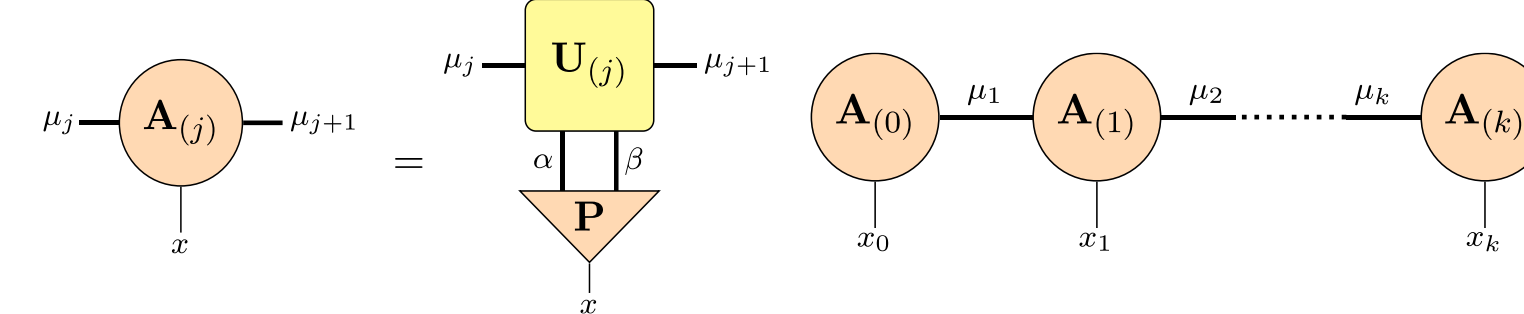
$$\Upsilon_{0:k}^{\text{SPP}} = \sum_{P_0, \dots, P_k} \text{Pr}(P_0, \dots, P_k) \bigotimes_{j=0}^k |P_j\rangle\langle P_j|.$$

- The multi-time Pauli twirl is operationally realized via randomized compiling.
- It removes genuinely quantum non-Markovianity while retaining classical spatiotemporal correlations, resulting in a **process-separable** quantum comb.

Efficient tensor network construction via local Pauli contractions



SPP probabilities form a classical TN



Virtual bonds carry environmental memory; physical legs label Pauli outcomes

Temporal correlation and HMMs

Transfer operators: quantify memory

$$T = \sum_x \mathbf{A}_x, \quad E_f = \mathbf{A}_f, \quad C_{f,g}(\tau) = \langle \mathbf{E}_f | T^\tau | \mathbf{E}_g \rangle$$

The transfer operator T propagates the latent memory (environment).

$$T = \sum_{i=1}^D \lambda_i |r_i\rangle\langle l_i|, \quad \xi = -\frac{1}{\ln |\lambda_2|}, \quad \Delta = 1 - |\lambda_2|$$

Small Δ / large $\xi \rightarrow$ slow correlation decay.

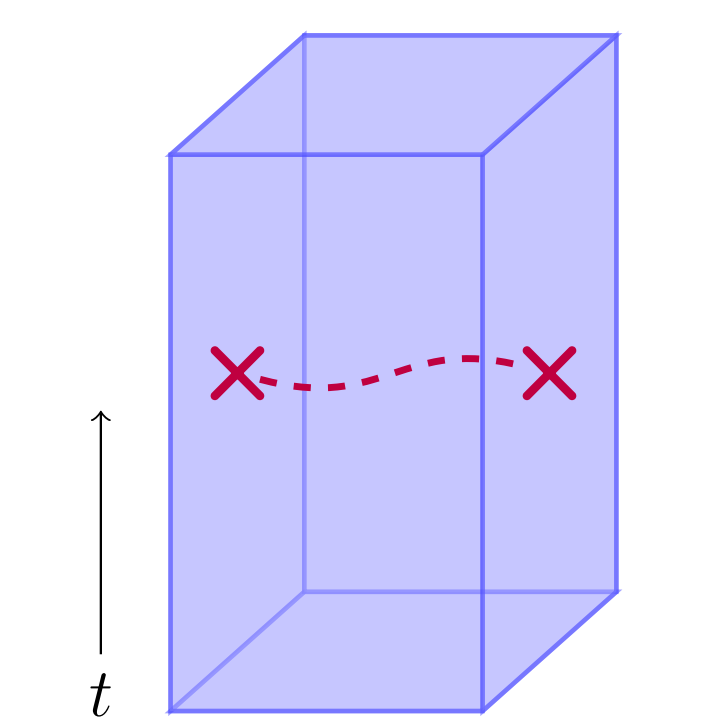
Hidden Markov models: sample memory

$$\text{Pr}(x_{0:k}) = \sum_{s_0, \dots, s_{k+1}} \pi_{s_0} \prod_{t=0}^k (K_{x_t})_{s_t s_{t+1}}.$$

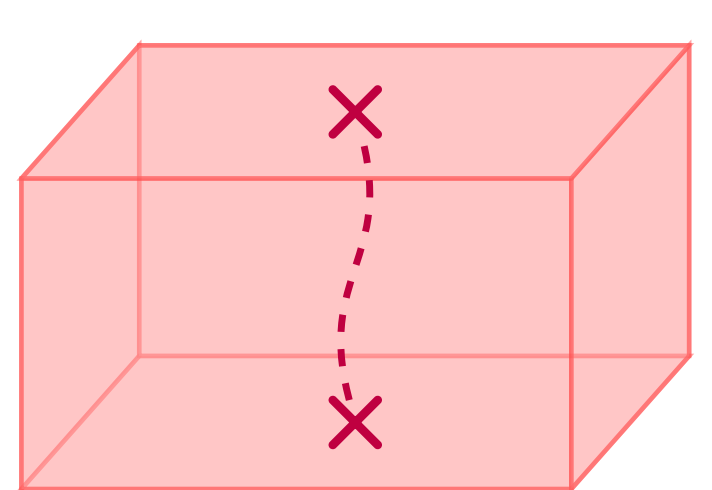
Stochastic, nonnegative SPP tensors define HMMs for fast correlated-fault sampling.

Application I: Tunable temporal correlations at fixed error rate

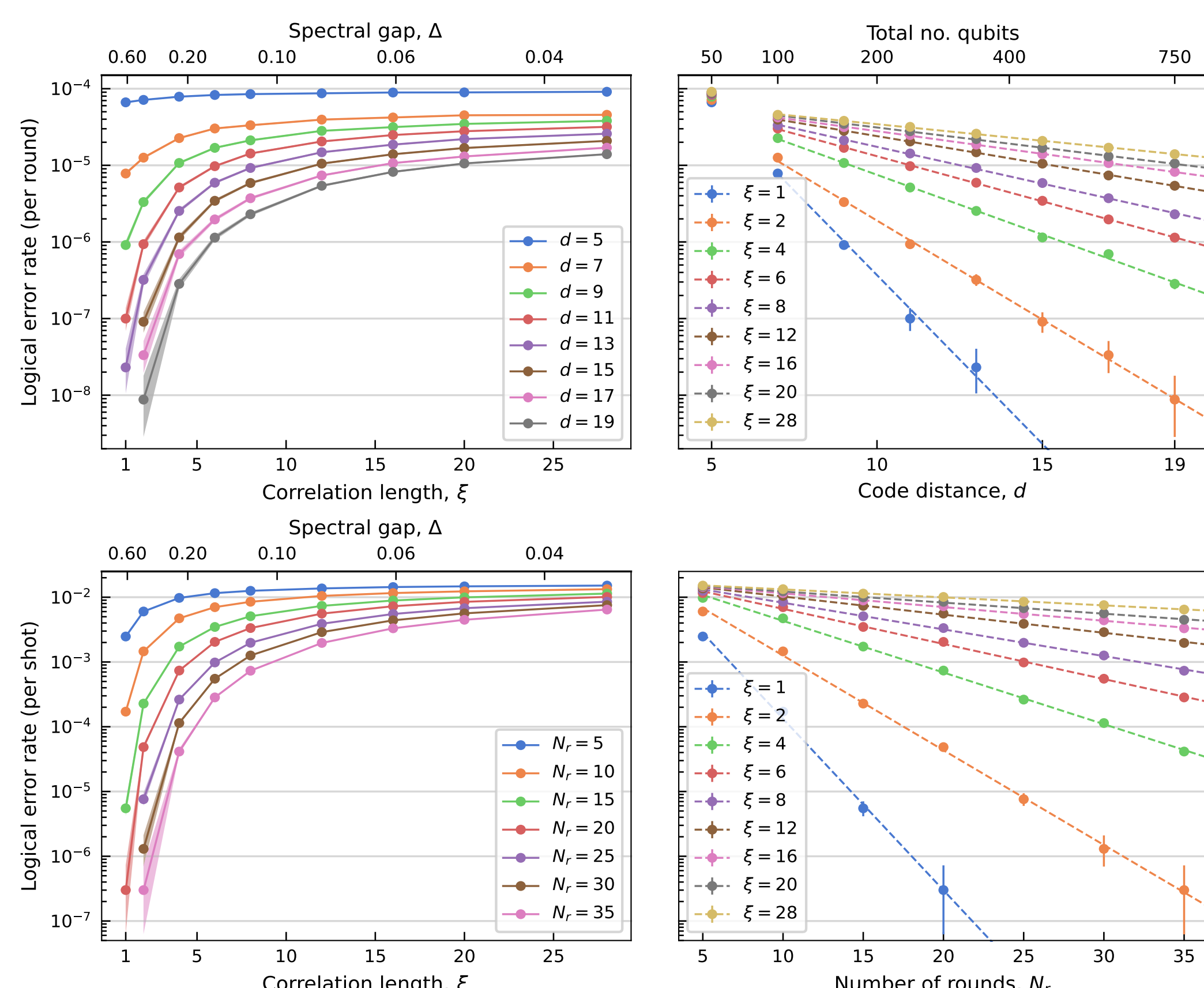
We instantiate an SPP as a two-state latent model, where qubits switch between *calm* and *storm* environments that emit Pauli faults. Tuning correlation time ξ at fixed marginal error rate shows that longer memory leads to damaging bursts that degrade surface code memory and stability.



Surface code memory



Surface code stability



Application II: Pseudo-critical spatiotemporal bath

We derive a 2D SPP from a coherent QCA bath; after system Pauli twirling, it maps exactly to a nonlinear probabilistic cellular automaton with $p_{\text{flip}} = \sin^2(k\theta)$. Near pseudo-criticality, enhanced density fluctuations and critical slowing down generate error avalanches that break distance scaling.

$$\mathcal{M}_{a,b,\theta} = \mathcal{U}^{SE} \circ \mathcal{Q}_{B \rightarrow R}^E(\theta) \circ \mathcal{Q}_{R \rightarrow B}^E(\theta) \circ \mathcal{S}^E(a, b)$$

